

Math 422: Time Series Analysis

Fall 2024

Course Description: Time series data is data that is sampled at several points in time. Such data is best analyzed with the tools of time series analysis. This course introduces numerous examples of, and exploratory data analysis techniques for, time series data. The course then covers the essential time series models, including ARIMA, SARIMA, Spectral Analysis, GARCH models, and the GLMM. We learn to fit these models via maximum likelihood estimation and numerical analysis techniques, how to verify the conditions required by each of the models, how to interpret the output of each model, and how to use the models for forecasting. After a detailed study of time series analysis, we will sample additional statistical modeling techniques in the latter half of the course. Weekly assignments demonstrate course topics via real-world applications, using the statistical computing language R to carry out the necessary computations. There will be a mid-semester project and a semester-long project where students can apply their knowledge to a data set of their choosing.

Class Details:

Instructor:	David White, david.white@denison.edu
Office:	Olin 207
Class Times:	MWF 3-3:50pm, Fellows 216
Drop-In Hours:	Mon 1:30-2:50, Wed 1:30-2:20, Fri 10:30-11:20 and by appointment , Olin 207
TA Study Sessions:	7-9pm Thurs; see Daily Plan document
Web Resources:	Canvas for HW and grades, Daily Plan Google Doc

For Remote Classes/Drop-In Hours (if we need any): Zoom Meeting ID: 926 8645 5656,

Passcode: math

<https://denison.zoom.us/j/92686455656?pwd=eEQrZldiOXNUbWwwOHZuNlJnZjRNUT09>

Daily plan: <https://tinyurl.com/mr3e335h>

GitHub: <https://github.com/denison-stats/math422fall2024>

TA: Thanh Do (do.t1@denison.edu). Please go to his TA sessions and ask lots of questions! You can also email him with questions, as he gets paid to answer emails, too!

Textbook: Time Series Analysis: A Data Analysis Approach Using R, by Robert H. Shumway and David S. Stoffer.

Course Learning Goals:

1. Practice how to break a complex problem into simple pieces. Develop your analytic thinking skills and problem-solving skills. Learn to mold theory-based approaches to solve messy real-world problems.
2. Learn how to read the textbook, how to figure out the big picture, and how to focus on the relevant details without being overwhelmed.
3. Learn to manipulate data sets in both R and Excel. Come to appreciate the power of simulation. Learn skills of data visualization, expectation, estimation, inference, and prediction
4. Develop group work skills, test-taking skills, and writing skills. Become proficient at re-writing and communicating technical content to a nontechnical audience.
5. Develop time management skills, metacognitive skills, and the habit of thinking intentionally about your learning and your goals.

Course Evaluation:

Labs and Quizzes: 48 %
Midsemester Project: 22 % Final Project 30%

Grading Scale: A standard 10% grading scale will be used. Therefore, 60% is required to pass the class, 70% will be a C-, 80% will be a B-, and 90% will be an A-.

Reading and videos: It is essential to read the book before class. To help you understand the book, I have recorded a video explaining each section, and you should watch this video before class. Furthermore, each day of class comes with a Daily R file, and you should also read that before class. These files are in the course GitHub and the Daily Plan Google doc tells you what to read and watch for each day. I run class meetings expecting you to have done this preliminary work, which allows us to dig into real-world examples, using R, during class. The goal of class will NOT be to review the reading. Instead, we'll apply the ideas in the reading to new problems. However, every class will begin with a chance for you to ask questions or seek clarification on the reading. Expect to spend 1.5-2 hours on reading and videos before each class meeting.

Laptop: Please bring your laptop to class every day, because you will need to do a Git-pull and compile Rmd files, so you will need certain packages to be required in RStudio. If you don't own a laptop, please contact me ASAP. Note that you can also use the Denison R server (r.denison.edu) if your computer has issues at any point. You should familiarize yourself with this resource before needing it in an emergency.

Labs: The bulk of your work in this class will be weekly labs in which you analyze real-world data sets in R, and write lab reports summarizing your main conclusions. I will provide R Markdown code to get you started. You will have a partner for each lab, and can work with this partner on all aspects of the lab, but should not be sharing code or answers with other students (though, you can ask them about R commands to do specific tasks, or to clarify material in the Daily R examples, upon which the labs are based). **When you write up your lab reports and large-scale projects, you must acknowledge any sources (including the internet, AI, the TA, and other students) that you used.** This is to instill lifelong habits in you of acknowledging sources consulted. It won't affect your grade to admit you got help. **No late labs will be accepted.** In consideration for sickness, personal emergencies, etc. I will drop the lowest lab grade. I am happy to answer questions, and I encourage you to come to my drop-in hours if you are confused about anything. You will get the most out of this time if you attempt the lab first and come with specific questions already prepared. Expect to spend at least 2-3 hours working on reading and labs outside of class for each of the four credit hours the course carries (i.e. 8-12 hours per week outside of class). This course adheres to Denison's Academic Credit Policy.

Quizzes: Some weeks, instead of a lab, we may have a quiz that would be worth the same as a lab. Each quiz will cover material presented in class, homework, and the assigned readings. The best way to study is to attend all class meetings and go through all the Daily Example files carefully. Quizzes will take the whole class period and will have an R portion, which is open-note. That means, you can use any R files on your computer, but not the internet or other students. The idea is to incentivize you to digest all the R materials. My philosophy is that the best way to learn is via the labs, but quizzes may be necessary if I detect that large portions of the class are not digesting the R materials provided.

Course Format. The course meets 3 days per week, for 50 minutes each day. Class MWF will begin with a Q & A forum where I will attempt to clear up any confusion you may have about the reading, video, or daily R file. Please take advantage of this and come with questions prepared. If no one has questions, then I'll do a super fast overview of the reading (i.e., a summary of the video summary) and then dive into the new examples in the daily R file. So if things are unclear, it's best to ask questions! Labs will be due on Fridays.

Communication: It cannot be stressed enough how essential communication is to succeeding in this course. After identifying topics that may be giving you trouble, please communicate this

information to me. There's no such thing as a bad or unwelcome question. Additionally, please communicate with each other. I view the class as a team trying to learn the material together. Collaboration will help all parties achieve this goal, as explaining concepts and examples to each other is a great way to learn. That said, don't share code or answers to lab questions, except with your lab partner.

Mid-semester project: Early in the semester you will obtain data on a topic you'll choose in consultation with me. This will be a data set on which you plan to fit both a 'function of time' model and an ARIMA model. This is similar to the final project from Math 220: you will write a paper discussing where the data came from, conducting your analysis, and reporting your conclusions. Your paper will be written in RMarkdown. You'll give a short presentation on it. I will give you guidelines for how to write it. This is a solo project and you must do the work yourself (but help from me or the TA is allowed).

Final Project: Halfway through the semester you will obtain data on a topic you'll choose in consultation with me. This will be a data set of more complexity than the mid-semester data set. Each time we learn something new you'll interpret it or test it on your pet data set, so it should exhibit sufficient complexity to do ARIMA, Spectral Analysis, GARCH (if residuals are heteroscedastic), and GLMM models. You'll come up with questions about this topic you hope to answer, and may need to obtain more data as the course progresses. At the end of the semester you'll hand in a longer paper answering your questions and give a presentation of your findings to the class. The paper will be due during the final exam slot. There will not be a final exam. You will give a presentation of your paper.

Oral Assessment: Part of your grade, for labs and these larger projects, will be based on your ability to explain how you did your data analysis. To prepare you for the kinds of questions you would face in a workplace (from your boss) or in a statistical consulting job (for your client), I will ask you questions like 'How did you deal with this outlier?' and 'Why did you choose an ARIMA instead of a SARIMA model?' Each week, after you hand in your lab (on Friday), I will randomly select some students to meet with me one-on-one to explain their lab to me (on Monday). This means that each individual student should have a complete command over the lab, even if it was done in a partnership. It also means that you would be crazy to rely on material from the internet or AI (even though this is explicitly forbidden anyway) because the oral explanation part will be impossible if you don't understand what you handed in. The same is true in the workplace. Get used to being able to explain your work, as that is what will distinguish you in life. I will arrange things so that all students do these oral explanation sessions equally often, probably 2-3 times. The goal is to teach you a valuable skill and I will try to not make these too stressful for you. There won't be 'gotcha' questions.

Disability: Any student who feels he/she may need an accommodation based on the impact of a disability should contact me privately as soon as possible to discuss his/her specific needs. I rely on the Academic Resource Center in 020 Higley to verify the need for reasonable accommodations based on documentation on file in that office.

Keys to Success

- Review the material from class the same day it is given. Find a way to attach this new knowledge to things you already understand. Study a bit every day rather than in bursts.
- Focus note-taking in class on these connections you made to your unique prior knowledge, not on material already in the book, handouts, or slides.
- Read the textbook slowly and carefully, at your desk, with a notebook nearby to write down questions and work out reasons for steps you don't understand. Pay particular attention to examples, and try to work them out yourself before reading the solution. Write down questions to ask in class.

- **Start labs early.** Give yourself time to get stumped and to get past these difficulties. Don't forget about the time you need for debugging your code. Labs go out on Fridays. You should start Saturday or Sunday and aim to finish by Wednesday.
- Keep a list of key definitions, statistical tests, assumptions, and R functions you frequently use, and commit them to memory throughout the course. Test your memory each week.

Participation and Attendance: Class meetings will be highly interactive and our goal will be to involve all participants. Attending class, answering questions, and asking questions is therefore essential. Because each day builds on the previous day, I expect you to attend every single class. There is no formal notion of an 'excused absence.' If you do miss class (e.g., you are sick), it will be your responsibility to get caught up; the first step is to get the notes from someone who was there, to try to understand what we covered, and to ask any remaining questions to me during office hours, or to ask the TA during the TA session. Any absence will be detrimental for the group as a whole, so please make your best effort to arrive on time for every class and to stay the whole class period. If you are sick, it's better to stay home rather than risk infecting other students (or the professor). If you miss more than three class meetings (the equivalent of a full week), I reserve the right to lower your final course grade. If you miss more than six class meetings, I reserve the right to give you a failing grade in the course. I will approach you first, if your lack of attendance is becoming a problem.

Academic Integrity: The students and faculty of Denison University and the Department of Mathematics are committed to academic integrity and will not tolerate any violation of this principle. Academic dishonesty is, in most cases, intellectual theft. It includes, but is not limited to, providing or receiving assistance in a manner not authorized by the instructor in the creation of work to be submitted for evaluation. This standard applies to all work ranging from daily homework assignments to major exams. As is indicated in Denison's Student Handbook, available through my.denison.edu, instructors must refer every act of academic dishonesty to the Associate Provost, and violations may result in failure in the course, suspension, or expulsion. (For further information, see the Academic Misconduct and Sanctions sections of the Student Handbook or Section VII.B. of the Faculty Handbook.)

I expect that you will all abide by the honor code in this course. Please do not use resources outside of me, your fellow students, the TA, and the textbook. On labs, you may work with your lab partner on all aspects, but you should not consult with other groups. You should never be looking at code or answer text from other groups, and if two groups hand in work that seems too close to identical, I may report this to the honor board. You may use Google or AI to find R commands or to debug your R code, but you are forbidden from uploading whole questions or data sets to AI, or using online resources to answer word problems (like, 'interpret this graph in real-world term; is military spending increasing or decreasing?'). Collaboration on quizzes and exams is not permitted, nor is the use of any online resource. All violations of the honor code will be reported.

Generative AI: When you submit work for a grade in this course, you represent it as your own intellectual product. You may not submit for a grade any content (e.g., ideas, text, code, images) that was generated, in whole or in part, by Generative Artificial Intelligence tools (including, but not limited to, ChatGPT and other large language models). Doing so in this course constitutes cheating under the Code of Academic Integrity.¹ However, it is permissible to use Generative AI (GenAI) for self-study, to look up definitions and theorems, to generate examples to help you understand a definition or theorem, to check your answers for typos, to find R commands to accomplish specific tasks (like, 'how do I make a scatterplot in R?'), or to debug your R code ('I got this error, what does it mean?'). You are not allowed to upload the data sets I give you or the lab questions, to any website or GenAI. Doing so is a violation of the honor code, of my

¹This language comes from the George Washington University Provost's office.

intellectual property (since I created these materials to teach you, not to help GenAI train on my work), and of the license in GitHub that you agree to when you ‘pull’ the course materials. It is literally illegal to upload course materials to GenAI or the internet. Similarly, you should not use GenAI to suggest models for your midsemester and final project, nor to clean your data set, nor to create R code to analyze it, nor to write the paper where you present your analyses. You may use GenAI to learn about R commands for specific tasks (you can use Google for this, too), to help identify bugs in code you wrote, and to make sense of R error messages. You can use it to spell-check or grammar-check your final paper, but you should not use it to suggest changes to the wording, and you should not include whole sentences or phrases suggested by GenAI. After all, I’m grading YOU not GenAI, and the feedback I give is for you to become a better writer.

Email: I will frequently contact you via email. Please check your email regularly. I will also check my email regularly, but often not after 6pm.

Appropriate Use of Course Materials: The materials distributed in this class, including the syllabus, exams, handouts, study aides, and in-class presentations, are copyrighted and hence will be governed by the provisions of U.S. copyright law. These materials are provided solely for the educational use of students enrolled in this course. You are not permitted to re-distribute them for purposes unapproved by the instructor; in particular you are not permitted to post course materials (e.g., files from GitHub) or your notes from lectures and discussions online, or to share them with individuals outside the class (including future students). Unauthorized uses of course materials will be considered academic misconduct as well as a breach of copyright law. If you have a friend who wants the course materials (e.g., for their research, or to help with a capstone project in DA) please tell them to contact me.

Multilingual Support: Students who use English in addition to other languages are welcome to use the resources available at the Multilingual Learning Office (MLO). The MLO includes Morayo Akinkugbe, PhD, the Assistant Director of Multilingual Programming and Support; Anna Adams, the English Language Support Specialist; and the student consultants who work with them. They are all trained and experienced in helping students address the different issues that arise when working in more than one language. If English is not your first or only language, please consider utilizing this resource, which is available to ALL Denison students. Dr. Akinkugbe, Ms. Adams, and the student consultants offer a variety of support for L2 students, including consulting with you about your written language (grammar, syntax, word-choices), developing strategies to manage your reading assignments, assisting with class conversation and presentations, and helping to devise ways to develop and effectively use all your skills in English. You can set up an appointment via <https://denisonuappointments.as.me/mlo>, or by emailing the Multilingual Learning Office directly at englishhelp@denison.edu.

A Heartfelt Welcome Back: I am very excited to meet you all, to be a support structure for you should you need it, and to help facilitate lots of new friendships. We may face challenges this semester (e.g., anxiety related to the election, or personal tragedy), but we can get through those together. I am committed to working with you so that our time together gives you joy, excitement, and beauty (really, stats can be beautiful, I promise) rather than anxiety. Please make time to take care of yourselves, to be a support network for others in the class, and please communicate with me frequently about what is and is not working for you. You all matter to me as human beings, regardless of whether the course material ‘clicks’ easily for you or if you struggle from time to time. It’s okay to struggle, and I am confident we can all grow together, in statistics, in our resilience, and as human beings.

Topics: A rough schedule of topics follows, but is subject to change. The schedule on the course web page supersedes the schedule here:

Week 1 = 1.1 - 1.3: Introduction to the course. First examples. Supplement on ‘functions of time models.’

Week 2 = 2.1 - 2.3: Time Related Dependence, Stationarity, autocorrelation.

Week 3 = 3.1 - 3.3: regression for time series, detrending, smoothing

Week 4 = functions of time models

Week 5 = 4.1 - 4.3: ARMA and ARIMA models, Unit Root Testing.

Week 6 = 4.4 - 5.2: estimation theory, forecasting.

Week 7 = 5.3 - 5.4: SARIMA, time series regression.

Week 8 = 6.1-6.3: GARCH, Spectral Analysis, filters

Week 9 = Midsemester Project Presentations.

Week 10 = 7.1-7.3: Periodogram, PCA

Week 11 = 7.4-8.2: Cross-Spectra, multivariate.

Week 12 = 8.3-8.5: long-memory, state space, threshold models

Week 13: Thanksgiving break

Week 14 = 8.6-8.7: bootstrapping, advanced topics, research

Week 15 = Final Project Presentations

Week 16: Final Project due during final exam slot.